CLAHE Preprocessing Experiment Report

Query-Side Contrast Enhancement for Nocturnal VPR

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Purpose

This experiment studies whether **CLAHE preprocessing** helps nocturnal visual place recognition in our project. The idea is simple: instead of changing the descriptors themselves, we preprocess the **query/night images** before feature extraction and compare the new runs against the existing no-preprocessing baselines.

Hypothesis

If campus night images are genuinely dark and low-contrast, then improving local contrast on the query side should reduce the day-night appearance gap and improve retrieval. We therefore expect CLAHE to help **more on campus** than on GardensPoint.

What Was Implemented

We added a shared preprocessing module in:

- `evaluation/preprocessing.py`

The shared evaluation scripts now support:

- `--preprocess none`
- `--preprocess clahe_query`
- `--preprocess clahe_all`
- `--clahe_clip_limit`
- `--clahe_tile_grid`

For this report we used `clahe_query`, because that matches the intended use case: improve the night/query images without altering the day map.

Experimental Setup

CLAHE settings used in all runs

- mode: `clahe_query`
- clip limit: `2.0`
- tile grid size: `8`

Descriptors tested

- **CosPlace**: strong VPR-specific baseline already used throughout the project
- **EigenPlaces**: another strong modern baseline that performs well on ranking
- **SALAD**: strongest practical modern descriptor in our recent experiments
- **SelaVPR++**: strongest benchmark descriptor overall in our recent experiments

Datasets / protocols tested

- GardensPoint benchmark
- campus strict image-level protocol
- campus place-level protocol

Commands Used

```
python demo.py --descriptor SALAD --dataset GardensPoint \
  --preprocess clahe_query --save_results

python test_campus_dataset.py --descriptor SALAD \
  --preprocess clahe_query --save_results

python test_campus_dataset_place_level.py --descriptor SALAD \
  --preprocess clahe_query --save_results

python demo.py --descriptor "SelaVPR++" --dataset GardensPoint \
  --preprocess clahe_query --save_results

python test_campus_dataset.py --descriptor "SelaVPR++" \
  --preprocess clahe_query --save_results

python test_campus_dataset_place_level.py --descriptor "SelaVPR++" \
  --preprocess clahe_query --save_results

python demo.py --descriptor CosPlace --dataset GardensPoint \
  --preprocess clahe_query --save_results

python test_campus_dataset.py --descriptor CosPlace \
  --preprocess clahe_query --save_results

python test_campus_dataset_place_level.py --descriptor CosPlace \
  --preprocess clahe_query --save_results
```

```
python demo.py --descriptor EigenPlaces --dataset GardensPoint \
  --preprocess clahe_query --save_results

python test_campus_dataset.py --descriptor EigenPlaces \
  --preprocess clahe_query --save_results

python test_campus_dataset_place_level.py --descriptor EigenPlaces \
  --preprocess clahe_query --save_results
```

Results

CosPlace

Setting	AUC	ΔAUC	R@100P	ΔR@100P	R@1	ΔR@1
GardensPoint baseline	0.592	–	0.270	–	0.350	–
GardensPoint + CLAHE	0.573	-0.019	0.225	-0.045	0.330	-0.020
Campus strict baseline	0.470	–	0.030	–	0.727	–
Campus strict + CLAHE	0.454	-0.016	0.061	+0.031	0.727	+0.000
Campus place baseline	0.627	–	0.114	–	0.969	–
Campus place + CLAHE	0.629	+0.002	0.110	-0.004	0.969	+0.000

EigenPlaces

Setting	AUC	ΔAUC	R@100P	ΔR@100P	R@1	ΔR@1
GardensPoint baseline	0.675	–	0.220	–	0.365	–
GardensPoint + CLAHE	0.655	-0.020	0.240	+0.020	0.355	-0.010
Campus strict baseline	0.443	–	0.000	–	0.788	–
Campus strict + CLAHE	0.488	+0.045	0.030	+0.030	0.879	+0.091
Campus place baseline	0.663	–	0.110	–	0.922	–
Campus place + CLAHE	0.663	+0.000	0.100	-0.010	0.938	+0.016

SALAD

Setting	AUC	ΔAUC	R@100P	ΔR@100P	R@1	ΔR@1
GardensPoint baseline	0.924	–	0.385	–	0.450	–
GardensPoint + CLAHE	0.896	-0.028	0.430	+0.045	0.415	-0.035
Campus strict baseline	0.504	–	0.030	–	0.818	–
Campus strict + CLAHE	0.522	+0.018	0.030	+0.000	0.848	+0.030
Campus place baseline	0.653	–	0.130	–	0.969	–
Campus place + CLAHE	0.651	-0.002	0.128	-0.002	0.969	+0.000

SelaVPR++

Setting	AUC	ΔAUC	R@100P	ΔR@100P	R@1	ΔR@1
GardensPoint baseline	0.937	–	0.355	–	0.465	–
GardensPoint + CLAHE	0.933	-0.004	0.495	+0.140	0.440	-0.025
Campus strict baseline	0.464	–	0.061	–	0.909	–
Campus strict + CLAHE	0.481	+0.017	0.061	+0.000	0.909	+0.000
Campus place baseline	0.658	–	0.158	–	0.969	–
Campus place + CLAHE	0.659	+0.001	0.152	-0.006	0.969	+0.000

Interpretation

1. CLAHE does *not* help uniformly

The effect is dataset-dependent and metric-dependent.

- On **GardensPoint**, CLAHE reduced AUC for *all four* descriptors and reduced R@1 for all four as well.
- On **campus strict**, CLAHE helped the stronger learned descriptors most clearly: EigenPlaces, SALAD, and SelaVPR++ all improved AUC, while EigenPlaces and SALAD also improved R@1.
- On **campus place-level**, CLAHE made only small changes overall, because the task is already less brittle once multiple valid views are credited.

2. This matches the EDA

Our raw-data EDA already suggested this outcome:

- GardensPoint night images appear **exposure-compensated / over-brightened**.
- Campus night images are **genuinely darker** and slightly lower contrast.

So CLAHE is more likely to help on campus, where there is a real contrast problem, than on GardensPoint, where the night images are already brightened.

3. Why did R@100P improve on GardensPoint?

For some descriptors, CLAHE improved **R@100P** on GardensPoint even though AUC and R@1 went down.

This suggests the following interpretation:

- CLAHE may slightly clean up the *thresholded* match distribution, making it easier to reach a perfect-precision operating point.
- But it also distorts some local appearance cues enough to hurt top-1 ranking.

So CLAHE may help the binary decision boundary in some cases while hurting the very best-match ordering.

4. Why is campus place-level almost unchanged?

The place-level protocol already gives credit for multiple valid views of the same place. That makes the task less brittle than the strict image-level protocol. Because of that, CLAHE has less room to help: the descriptor is already solving most of the task once multi-view positives are allowed.

5. Best practical conclusion

The strongest practical conclusion from these runs is:

- **Do not treat CLAHE as a universal default.**
- It is **most promising for the strict campus setup**, where the query images are truly dark and the exact image-level matching protocol is sensitive to appearance shifts.
- It is **not clearly useful on GardensPoint**, and may actually hurt top-1 retrieval there.

6. Descriptor-specific summary

- **CosPlace**: CLAHE does not meaningfully help. It slightly hurts GardensPoint and strict-campus AUC, while leaving R@1 unchanged on campus.
- **EigenPlaces**: this is the clearest positive baseline result. CLAHE noticeably improves strict-campus AUC and R@1, while leaving place-level AUC unchanged.
- **SALAD**: CLAHE helps the strict campus setup, but hurts GardensPoint overall.
- **SelaVPR++**: CLAHE gives a small strict-campus AUC gain, but mostly leaves the already-strong place-level ranking unchanged.

Qualitative Note

The strict campus CLAHE runs reduced the number of false positives while keeping or improving top-k retrieval:

- EigenPlaces strict: AUC rose from 0.443 to 0.488 and R@1 rose from 0.788 to 0.879
- SALAD strict: FP fell from 6 to 4, and R@1 rose from 0.818 to 0.848
- SelaVPR++ strict: FP stayed at 2, AUC rose from 0.464 to 0.481, while R@1 stayed at 0.909

This is a good sign that CLAHE may be more useful as a **campus-specific query enhancement** than as a general benchmark improvement.

Run Artifacts

The new CLAHE runs are saved under:

- `output_images/runs/benchmarks/0015_demo_gardenspoint_salad_clahe_query_...`
- `output_images/runs/campus_strict/0008_campus_strict_salad_clahe_query_...`
- `output_images/runs/campus_place_level/0007_campus_place_level_salad_clahe_query_...`
- `output_images/runs/benchmarks/0016_demo_gardenspoint_selavpr_clahe_query_...`
- `output_images/runs/campus_strict/0009_campus_strict_selavpr_clahe_query_...`
- `output_images/runs/campus_place_level/0008_campus_place_level_selavpr_clahe_query_...`
- `output_images/runs/benchmarks/0017_demo_gardenspoint_cosplace_clahe_query_...`
- `output_images/runs/campus_strict/0010_campus_strict_cosplace_clahe_query_...`
- `output_images/runs/campus_place_level/0009_campus_place_level_cosplace_clahe_query_...`
- `output_images/runs/benchmarks/0018_demo_gardenspoint_eigenplaces_clahe_query_...`
- `output_images/runs/campus_strict/0011_campus_strict_eigenplaces_clahe_query_...`
- `output_images/runs/campus_place_level/0010_campus_place_level_eigenplaces_clahe_query_...`

Conclusion

CLAHE is worth keeping in the project as a configurable option, but the experiments show that its usefulness is **domain-specific**. It helps most where the night images are truly dark and contrast-limited, which is exactly what our campus EDA suggested. The cleanest next step is therefore to keep CLAHE as an optional preprocessing branch for **campus-style deployment**, not as a universal preprocessing default for every benchmark.